ClipSAT : Math for AI & Data Science

Unit 2.01: Neural Layer Matrices & Activations

Name: _____

Q1. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{50} to mathbf{y} in mathbb{R}^{25} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[ ] A: 1275
[ ] B: 1250
[ ] C: 75
[ ] D: 1325

Q2. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{60} to mathbf{y} in mathbb{R}^{30} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[ ] A: 1830
[ ] B: 1800
[ ] C: 90
[ ] D: 1890

Q3. Evaluate the output of the standard ReLU activation function fz = max0, z for inputs z_1 = -4 and z_2 = 6 .

[ ] A: fz_1 = 4, ; fz_2 = 6
[ ] B: fz_1 = -4, ; fz_2 = 6
[ ] C: fz_1 = 0, ; fz_2 = 0
[ ] D: fz_1 = 0, ; fz_2 = 6

Q4. Evaluate the output of the standard ReLU activation function fz = max0, z for inputs z_1 = -6 and z_2 = 9 .

[ ] A: fz_1 = 6, ; fz_2 = 9
[ ] B: fz_1 = -6, ; fz_2 = 9
[ ] C: fz_1 = 0, ; fz_2 = 0
[ ] D: fz_1 = 0, ; fz_2 = 9

Q5. Evaluate the output of the standard ReLU activation function fz = max0, z for inputs z_1 = -10 and z_2 = 15 .

[ ] A: fz_1 = 0, ; fz_2 = 0
[ ] B: fz_1 = -10, ; fz_2 = 15
[ ] C: fz_1 = 0, ; fz_2 = 15
[ ] D: fz_1 = 10, ; fz_2 = 15

Q6. Evaluate the output of the standard ReLU activation function fz = max0, z for inputs z_1 = -12 and z_2 = 18 .

[ ] A: fz_1 = 0, ; fz_2 = 0
[ ] B: fz_1 = -12, ; fz_2 = 18
[ ] C: fz_1 = 0, ; fz_2 = 18
[ ] D: fz_1 = 12, ; fz_2 = 18

Q7. The sigmoid activation function is sigmaz = frac{1}{1 + e^{-z}} . If sigmaz_0 = 0.20 , what is the value of the derivative sigma'z_0 ?

[ ] A: 0.1600
[ ] B: 0.4000
[ ] C: 0.8000
[ ] D: 0.0400

Q8. The sigmoid activation function is sigmaz = frac{1}{1 + e^{-z}} . If sigmaz_0 = 0.40 , what is the value of the derivative sigma'z_0 ?

[ ] A: 0.2400

[ ] B: 0.8000

[ ] C: 0.6000

[ ] D: 0.1600